

SHUBHANGI KATARIYAR

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📁 [shubhangikatarivar](https://github.com/shubhangikatarivar)

🌐 shubhangikatariyar.github.io/studio

Education

University of Wisconsin – Madison

Aug 2026 – Dec 2027

Masters in Electrical & Computer Engineering - Machine Learning & Signal Processing

Madison, WI

University of Mumbai

Aug 2018 – May 2022

Bachelor of Engineering in Electronics & Telecommunications GPA: 3.7/4.0

Mumbai, India

Professional Experience

Reserve Bank Information Technology

Nov 2025 – Jun 2026

Data Scientist

Navi Mumbai, India

- Built a production-grade **Agentic AI Chatbot** adopted by **100+ business users**, automating analytical query generation from natural language and reducing time-to-insight across enterprise datasets.
- Architected a **low-code GenAI platform** for creating **RAG-powered knowledge repositories** from **100K+ enterprise documents**, enabling rapid deployment of AI applications across business functions.
- Engineered advanced **retrieval pipelines** using **Entity-Based**, **Section-Wise**, and **Research-Driven chunking strategies**, improving retrieval relevance and response grounding for enterprise LLM systems.
- Led a team of **8 engineers** in delivering AI solutions spanning **document intelligence**, **OCR**, **invoice automation**, and **anomaly detection**, supporting **20+ enterprise business workflows**.

Tata Consultancy Services

Jun 2022 – Oct 2025

AI Engineer

Mumbai, India

- Standardized responsible AI practices across 20+ enterprise projects by contributing to TCS's ethical assurance platform, now adopted as an org-wide governance framework.
- Designed a bias-detection system in **Python** to audit datasets/models for **statistical parity**, **equalized odds**, and **disparate impact**, reducing discrimination risks by 25% in loan-approval systems.
- Achieved **86% detection accuracy** and reduced **LLM hallucinations by 21%** by developing a **RAG-based validation pipeline**, improving the factual consistency and reliability of AI-generated responses.
- Designed and deployed an interactive **Explainable AI dashboard** leveraging **SHAP** and **ALE**, reducing manual credit risk review effort by 30% and enabling adoption in high-risk AI governance workflows.
- Accelerated **ML model deployment** by 60% through automated **MLOps pipelines**, enabling seamless orchestration of data processing, model training, testing, and production releases.
- Improved **object detection reliability** by 38% through confidence calibration using **logistic regression** and **Grad-CAM** visualizations, increasing prediction trustworthiness in quality assurance systems.

Tata Consultancy Services

Mar 2022 – May 2022

NLP Project Intern

Mumbai, India

- Achieved **92% classification accuracy** and an **F1 score of 0.89** by developing a **BERT-based transformer model** to categorize 10,000+ patents using abstract and CPC data from the USPTO API.
- Accelerated preprocessing pipeline for patent text classification by implementing **tokenization**, **stemming**, and **TF-IDF**, enabling scalable feature extraction for deep learning models.

Projects

Cine AI: Content-Based Movie Recommender ([Demo](#)) | *Python, Streamlit, Scikit-learn, Heroku*

- Engineered a content-based recommender using cosine similarity and count vectorization, processing TMDB metadata to deliver personalized movie suggestions.
- Deployed an interactive web application using Streamlit and Heroku, achieving 85%+ user satisfaction during testing.

Renal Cell Carcinoma Segmentation & Survival Prediction ([IEEE Xplore](#)) | *Python, U-Net, CECT Imaging*

- Achieved **84.02% segmentation accuracy** and **0.80 Dice score** using a **U-Net** for tumor detection in **CECT scans**, enabling precise boundary identification.
- Published in **ICAST 2023 – IEEE Xplore** with a novel **survival prediction model** analyzing tumor heterogeneity.

Technical Skills

Programming Languages: Python (Pandas, NumPy, SciPy, spaCy, Matplotlib), C++, SQL, NoSQL, Shell Scripting

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face, Computer Vision, NLP, DL

Generative AI: RAG, Agentic AI, LangChain, LangGraph, Multi-Agent Systems, MCP, LLM Fine-Tuning, OpenAI API, Prompt Engineering, LoRA, PEFT, Quantization, Pruning, Distillation

MLOps & Deployment: Jenkins, Docker, Kubeflow, TFX, MLflow, Airflow, Kafka, Spark, REST APIs, OpenShift (OCP)

Cloud, Data & Developer Tools: AWS, Azure, GCP, MongoDB, Hadoop, Git, Postman, Tableau, MobaXterm, VS Code